

ANARCHY

AI Platform

Top 50 Problems + Top 100 Improvements

Technical Audit Report — File 7 of 8

50	12	100	9.9
Total Problems	Critical Issues	Improvements	Achievable Score

Top 50 Problems — Ranked by Severity & Impact

Complete prioritized issue registry across all audit categories

#	Severity	Category	Problem	Impact	Effort
#01	CRITICAL	Architecture	TreeFlowCanvas.tsx — 2000+ line God Component	Unmaintainable; single point of failure for canvas, AI, history, export	3 - 4 days
#02	CRITICAL	Security	API Key in Renderer Process (SEC-001)	Replicate key visible in DevTools — data breach risk	2 days
#03	CRITICAL	Security	Supabase RLS Incomplete (SEC-002)	Cross-user data leakage on all history, library, canvas tables	1 day
#04	CRITICAL	Security	Unauthenticated Local HTTP Bridge (SEC-003)	Any local process can inject malicious images into canvas	1 day
#05	CRITICAL	Architecture	No AI Job Queue in ReplicateService	5 concurrent requests cascade-fail; production unusable under load	1 week
#06	CRITICAL	Architecture	Zero Test Coverage	No unit, integration, or E2E tests — regressions undetectable	2 weeks
#07	CRITICAL	Functional	No React Error Boundaries on Canvas	Single node failure causes full white screen with no recovery	0.5 day

#	Severity	Category	Problem	Impact	E ff o r t
#08	CRITICAL	Performance	History Page Memory Exhaustion	All records loaded into RAM — crash on 500+ records	2 d a y s
#09	CRITICAL	Performance	Re-render Storm — 30+ per user action	Broad Zustand subscriptions cause mass unnecessary re-renders	3 d a y s
#10	CRITICAL	Architecture	LiteGraph Migration 40% Complete	Hybrid ReactFlow+LiteGraph causes race conditions and state bugs	1 w e e k
#11	CRITICAL	Functional	No Canvas Viewport Culling	All canvas nodes rendered even when outside viewport	2 d a y s
#12	CRITICAL	Functional	Double-Submit on Generate Button	Button not disabled during active job — duplicate credits charged	0. 5 d a y
#13	HIGH	Security	Stripe Webhook Not Verified (SEC-004)	Replay attack can add unlimited credits without payment	0. 5 d a y
#14	HIGH	Security	Credit Overdraft Race Condition (SEC-005)	Rapid concurrent generations bypass credit balance check	1 d a y
#15	HIGH	Performance	ReactFlow Bundle Dead Weight	~120KB removed after LiteGraph migration complete	0. 5 d a y
#16	HIGH	Functional	No Generation Cancel Button	User trapped in 30-60s wait with no exit	1 d a y

#	Severity	Category	Problem	Impact	Effort
#17	HIGH	Performance	No Code Splitting on Routes	Entire 830KB bundle loads synchronously on startup	1 day
#18	HIGH	UX	Fake Generation Progress Timer	Erodes trust — 73% indicator not reflecting real progress	1 day
#19	HIGH	Functional	Nano Banana 2 Seed Confusion	Seed UI shown for model that ignores seeds — variations look identical	0.5 day
#20	HIGH	Architecture	GhostNode.tsx — 600+ line Multi-type Component	10 node types in one component — unmaintainable, blocks LiteGraph migration	2-3 days
#21	HIGH	Data	No History Pagination	Flat load of all records — power user crash scenario	2 days
#22	HIGH	Security	Prompt Injection via Node Labels (SEC-006)	Shared canvas files can inject into AI prompts	1 day
#23	HIGH	Performance	Image Object URLs Never Revoked	Blob URLs accumulate all session — 50MB+ memory leak per session	0.5 day
#24	HIGH	Security	No Content Security Policy (SEC-007)	Injected scripts run freely in renderer	1 day
#25	HIGH	Functional	DXF Bridge — No Health Check	Silent failures when Python process crashes	1 day

#	Severity	Category	Problem	Impact	Effort
#26	HIGH	UX	Generic Error Messages	'Generation failed' with no cause or actionable next step	1 day
#27	HIGH	Architecture	HistoryService.ts — 6 Responsibilities	God Service — CRUD + search + pagination + thumbnail + export	2 days
#28	HIGH	Architecture	ExportService — Sync I/O in Renderer	File writes block Electron UI thread during export	2 days
#29	HIGH	UX	No Onboarding / First-Run Experience	Blank canvas on first launch — highest churn point	1 week
#30	HIGH	Performance	Store Writes During Pan/Zoom Animation	Canvas viewport position saved on every mousemove event	0.5 day
#31	HIGH	Performance	Full-res Images as Canvas Thumbnails	4K renders used as 240px node previews — 8MB+ per node	1 day
#32	HIGH	Functional	History Search — Client-side Linear Scan	O(n) search on every keystroke — freezes on 500+ records	1 day
#33	HIGH	UX	No Keyboard Shortcuts for Core Actions	No Ctrl+Z, Ctrl+Enter, Space+drag — feels like amateur software	1 day
#34	HIGH	UX	No Node Palette — GhostNode Types Hidden	Users cannot discover available node types	1 day

#	Severity	Category	Problem	Impact	Effort
#35	MEDIUM	UX	No Canvas Minimap	Large workflows are impossible to navigate without minimap	2 days
#36	MEDIUM	Architecture	No Zustand Middleware	No devtools, no immer, no persist — hard to debug state	0.5 day
#37	MEDIUM	UX	No Canvas Workflow Templates	Blank canvas intimidating — no Quick Start	3 days
#38	MEDIUM	Functional	Library — No Organization	Flat image list unusable for professional multi-project use	1 week
#39	MEDIUM	Security	No Rate Limiting on Generation	User can fire 100 generations/min, depleting credits	1 day
#40	MEDIUM	Performance	No Lazy Loading on History Thumbnails	Thumbnails load synchronously during page mount	2 days
#41	MEDIUM	UX	No Notification History	Dismissed toasts gone forever — no way to review past alerts	1 day
#42	MEDIUM	Performance	No Supabase Query Indexes	Missing composite index on (user_id, created_at)	0.5 day
#43	MEDIUM	UX	No Global Search	Must navigate to each section to search — no Cmd+K	2 days

#	Severity	Category	Problem	Impact	Effort
#44	MEDIUM	Architecture	AIConfigStore — API Key in Zustand	Credentials stored in plain state — accessible via DevTools	1 day
#45	MEDIUM	Functional	PDF Export — In-memory All Pages	OOM risk on large multi-page exports	2 days
#46	MEDIUM	Security	No Account Lockout on Failed Login	Brute force login attempts possible	0.5 day
#47	MEDIUM	UX	No Empty State Illustrations	History/Library/Projects pages show blank — feel unfinished	1 day
#48	MEDIUM	Architecture	No Documentation / JSDoc	Zero inline documentation — knowledge siloed in one developer	2 weeks
#49	LOW	Security	Electron App Not Code-Signed	Windows SmartScreen warning on install	3 days
#50	LOW	UX	No Side-by-Side Upscale Comparison	No before/after slider for upscale quality review	2 days

Top 100 Improvements

Complete improvement backlog sorted by sprint and impact

Sprint 1 — Critical Fixes (2 weeks)

01	Add React Error Boundary around entire canvas + each node type
02	Disable Generate button + show spinner during active AI job
03	Move Replicate API key to Electron main process (safeStorage + IPC)
04	Enable Supabase RLS on history, library_images, canvas_states, notifications tables
05	Add shared secret token authentication to local HTTP bridges (port 14400 + 7430)
06	Add pre-flight credit check before every generation request
07	Fix Stripe webhook signature verification (constructEvent)
08	Replace credit check-then-deduct with atomic Supabase RPC
09	Add generation cancel button with AbortController
10	Add real Replicate progress steps (uploading → processing → downloading)
11	Add DXF bridge health check + 10-second timeout with clear error message
12	Show Nano Banana 2 capability badge: 'Seeds: Not Supported'

Sprint 2 — Architecture Refactor (3 weeks)

01	Split TreeFlowCanvas.tsx into: CanvasEngine, NodeManager, useAIOrchestrator, useCanvasPersistence, CanvasContextMenu, GhostNodeOrchestrator, CanvasToolbar
02	Migrate all 10 GhostNode types to LiteGraph custom node classes
03	Create GhostNodeRegistry for dynamic node type registration
04	Remove ReactFlow dependency entirely after migration
05	Implement AIJobQueue (p-queue with concurrency 3) in Electron main process
06	Add exponential backoff retry to ReplicateService (3 attempts, 1s/2s/4s)
07	Add circuit breaker to ReplicateService (open after 5 failures)
08	Split ReplicateService into: FluxService, NanoBananaService, UpscaleService
09	Add granular Zustand selectors on all canvas components
10	Wrap all canvas node components in React.memo with custom comparators
11	Add Zustand devtools + immer + persist middleware to all stores
12	Move API key from AIConfigStore to Electron safeStorage

Sprint 3 — Performance Sprint (2 weeks)

01	Implement cursor-based pagination in HistoryService + HistoryPage
02	Add virtual scroll to HistoryPage (react-virtualized or Tanstack Virtual)
03	Move History search to Supabase full-text search (pg_trgm)
04	Add URL.revokeObjectURL() after every image loads into canvas
05	Debounce canvas viewport store writes to 100ms idle threshold
06	Enable LiteGraph dirty region tracking (setDirty per changed node)
07	Add viewport culling: skip draw() for nodes outside visible bounds
08	Resize images to 2x display size before injecting as canvas thumbnails
09	Add React.lazy() + Suspense for all page routes
10	Add Vite manualChunks for vendor / canvas / app separation
11	Create local image cache in Electron userData folder
12	Add Supabase composite index on (user_id, created_at DESC)

Sprint 4 — UX Sprint (2 weeks)

01	Add actionable error messages for all failure modes (generation, export, auth, billing)
02	Add floating node palette (Tab key) showing all 10 GhostNode types with previews
03	Add canvas minimap (M key toggle, bottom-right corner)
04	Add 3 workflow templates on new canvas: Quick Render, Variation Explorer, Day/Night
05	Add Cmd+K global spotlight search across history + library + canvas
06	Add 'Load in Canvas' button on every history card
07	Add skeleton card loaders for History and Library pages
08	Add notification deduplication (same message within 5s → count badge)
09	Add notification history bell icon in header (last 50 notifications)
10	Add Ctrl+Z undo / Ctrl+Y redo / Ctrl+Enter generate / Space+drag pan
11	Add ? shortcut overlay modal with full keyboard reference
12	Add empty state illustrations for History, Library, Projects pages

Sprint 5 — Onboarding + Polish (2 weeks)

01	Create Quick Start sample project that opens on first launch
----	--

02	Add 5-step interactive canvas tutorial (skip-able)
03	Add API key setup wizard with step-by-step guidance
04	Add contextual help ? tooltip on every major feature
05	Add 'Discover' tab with 3 prompt ideas for common arch scenarios
06	Add credit cost display before generation ($\approx$ N credits)
07	Add prompt history dropdown (last 10 prompts)
08	Add negative prompt field to generation panel
09	Add batch generation (4 seeds in one click)
10	Add canvas node grouping with colored frame + collapse
11	Add side-by-side upscale comparison (before/after slider)
12	Add PNG + WebP export format options

Sprint 6 — Security + Reliability (2 weeks)

01	Add Content Security Policy to all Electron BrowserWindows
02	Add prompt sanitization + length limits on all user-controlled inputs
03	Add rate limiting: 10 generations/minute per user via Supabase Edge Function
04	Add billing_audit_log table (immutable append-only)
05	Fix DXF subprocess to use array args (no shell injection)
06	Add Supabase Auth failed attempt lockout + CAPTCHA
07	Add security event logging (failed logins, credit anomalies)
08	Run npm audit + remediate all HIGH/CRITICAL dependency vulnerabilities
09	Add streaming PDF generation (page-by-page, no in-memory buffer)
10	Add export queue (prevent concurrent file writes)
11	Add batch export: ZIP of all canvas outputs
12	Add export progress bar with page count

Sprint 7 — Library + Collaboration (3 weeks)

01	Add Library Collections / project folders
02	Add Library tag system with color coding
03	Enable drag-from-Library to Canvas as source node
04	Add image metadata view (model, prompt, date, size)
05	Add Library search (Supabase FTS on filename + prompt)

06	Add bulk upload with real progress indicators
07	Add Supabase CDN transform for auto image compression
08	Add Google OAuth social login via Supabase Auth
09	Add project sharing: generate shareable canvas link
10	Add canvas comment / annotation cards
11	Add auto-layout (arrange nodes by data flow direction)
12	Add node collapse/expand for complex canvases

Sprint 8 — Advanced Features (4 weeks)

01	Implement full AI Job Queue with UI status (queue position, ETA)
02	Add real-time Replicate webhook callback for progress (replace polling)
03	Add canvas workflow export/import (.ana file format)
04	Add Revit bridge integration (Revit → Anarchy AI source node)
05	Add local AI support via Ollama (local node type)
06	Add ZainCash payment integration for Iraqi market
07	Add MFA (TOTP) via Supabase Auth
08	Add Sentry performance monitoring + error tracking
09	Add custom FPS counter overlay in dev mode
10	Implement EV code signing for Windows .exe
11	Add auto-updater with signature verification
12	Set up Playwright E2E tests for all critical user flows

Sprint 9 — Plugin Architecture + 9.9/10 (ongoing)

01	Design and implement plugin API: NodeTypePlugin interface
02	Port all GhostNode types to plugin system
03	Create plugin marketplace skeleton (local plugins only first)
04	Add WebGL canvas renderer for 500+ node canvases
05	Add Level-of-Detail rendering (simplified nodes at low zoom)
06	Add real-time collaboration via Supabase Realtime (cursor sharing)
07	Add project version history with visual diff between canvas versions
08	Add AI-powered prompt suggestions (LLM analyzes reference image + suggests prompts)
09	Add client presentation mode (full-screen, hide UI chrome)

10	Achieve 80%+ test coverage (Vitest unit + Playwright E2E)
11	Complete external security penetration test
12	Launch public beta with 100-user waitlist

End of File 7 / 8

This concludes the Top 50 Problems + Top 100 Improvements registry. Continue with **File 8 — Roadmap + Competitor Analysis + Future Architecture** for the 1-year and 3-year strategic roadmap, comparison with Midjourney and Leonardo AI, and the future architecture vision (Job System, Event Bus, Plugin Architecture, Canvas Engine).